PREDICTIVE MODELING AND RISK SCORING FOR BANK CUSTOMER CHURN

Submitted By

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Internship Project Report

ABSTRACT

Customer churn is one of the most significant challenges faced by banks and financial institutions. When customers discontinue using banking services, organizations experience revenue loss, reduced customer lifetime value, and decreased opportunities for cross-selling and upselling. Therefore, identifying customers who are likely to leave the bank has become an important business objective.

This project focuses on developing a machine learning-based customer churn prediction system using customer demographic and banking information. The project includes data preprocessing, exploratory data analysis, feature engineering, model development, evaluation, and churn risk scoring. Multiple machine learning techniques were applied, including Logistic Regression and Random Forest. The Random Forest model achieved the highest accuracy of 86.45%, making it the final selected model.

The developed solution can help banks identify high-risk customers and implement proactive retention strategies to improve customer satisfaction and profitability.

Keywords: Customer Churn, Machine Learning, Random Forest, Predictive Analytics, Banking Analytics, Customer Retention.

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1. INTRODUCTION

The banking industry is becoming increasingly competitive. Customers have access to multiple financial institutions and can easily switch banks if they are dissatisfied with services, pricing, or customer support.

Customer churn refers to the process where customers stop using a company's products or services. High churn rates negatively impact revenue growth and increase customer acquisition costs. Therefore, predicting customer churn before it occurs has become a critical business requirement.

Machine Learning provides an effective solution by analyzing historical customer behavior and identifying patterns associated with churn. By predicting which customers are likely to leave, banks can take preventive actions such as personalized offers, loyalty programs, and targeted marketing campaigns.

This project aims to develop a predictive system capable of identifying customers at risk of leaving the bank.

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2. PROBLEM STATEMENT

Banks possess large volumes of customer data but often lack advanced analytical systems capable of predicting churn accurately.

Without predictive analytics:

• Retention efforts become reactive rather than proactive.

• Resources are wasted on customers who may not require intervention.

• High-risk customers are not identified in time.

The objective of this project is to use machine learning techniques to predict customer churn and generate churn risk scores that support effective customer retention strategies.

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3. OBJECTIVES

Primary Objectives

• Predict customer churn with high accuracy.

• Generate churn probability scores.

• Identify key churn drivers.

Secondary Objectives

• Improve customer retention.

• Support business decision-making.

• Reduce customer acquisition costs.

• Enable proactive engagement strategies.

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4. DATASET DESCRIPTION

The dataset contains customer information collected from a European banking institution.

Dataset Characteristics

Total Records: 10,000

Total Features: 14

Target Variable: Exited

Where:

0 = Customer Retained

1 = Customer Churned

Feature Description Table

Feature	**Description**
CreditScore	Customer credit score
Geography	Customer location
Gender	Customer gender
Age	Customer age
Balance	Account balance
Exited	Churn status

	Year	**CustomerId**	**Surname**	**CreditScore**	**Geography**	**Gender**	**Age**	**Tenure**	**Balance**	**NumOfProducts**	**HasCrCard**	**IsActiveMember**	**EstimatedSalary**	**Exited**
0	2025	15634602	Hargrave	619	France	Female	42	2	0.00	1	1	1	101348.88	1
1	2025	15647311	Hill	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0
2	2025	15619304	Onio	502	France	Female	42	8	159660.80	3	1	0	113931.57	1
3	2025	15701354	Boni	699	France	Female	39	1	0.00	2	0	0	93826.63	0
4	2025	15737888	Mitchell	850	Spain	Female	43	2	125510.82	1	1	1	79084.10	0

Figure 1: Preview of European Bank Customer Dataset.

The dataset contains customer demographic, financial, and banking information. It includes variables such as Credit Score, Geography, Gender, Age, Balance,Number of Products, Estimated Salary, and Churn Status (Exited).The dataset consists of 10,000 customer records and14 attributes.

Observation

The dataset contains demographic, financial, and behavioral information suitable for predictive analytics.

5. DATA PREPROCESSING

Data preprocessing was performed to improve data quality and prepare the dataset for machine learning.

Steps Performed

Missing Value Analysis

The dataset was checked for missing values.

Result:

No missing values were found.

	0
Year	0
CustomerId	0
Surname	0
CreditScore	0
Geography	0
Gender	0
Age	0
Tenure	0
Balance	0
NumOfProducts	0
HasCrCard	0
IsActiveMember	0
EstimatedSalary	0
Exited	0

dtype: int64

Figure 2: Missing Value Analysis of the Dataset.

Before performing exploratory data analysis and machine learning modeling,

the dataset was examined for missing or null values.

A missing value analysis was conducted using the Pandas isnull().sum()

function to identify incomplete records across all variables.

The results show that every attribute in the dataset contains zero missing values. Variables such as CreditScore, Geography, Gender, Age, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary, and Exited are completely populated.

Since no missing values were detected, no data imputation, deletion, or replacement techniques were required. This ensured that the dataset maintained its integrity and was suitable for further analysis and model development.

The absence of missing values improves the reliability of statistical analysis and contributes to better predictive performance of machine learning algorithms.

Observation:

The dataset is complete and contains no missing values.Therefore, no additional data cleaning was required before proceeding to exploratory data analysis and model training.

Removal of Unnecessary Features

The following variables were removed:

• CustomerId

• Surname

• Year

These features do not contribute significantly to churn prediction.

Encoding Categorical Variables

Categorical variables such as Geography and Gender were converted into numerical form using One-Hot Encoding.

	CreditScore	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited	Geography_Germany	Geography_Spain	Gender_Male
0	619	42	2	0.00	1	1	1	101348.88	1	False	False	False
1	608	41	1	83807.86	1	0	1	112542.58	0	False	True	False
2	502	42	8	159660.80	3	1	0	113931.57	1	False	False	False
3	699	39	1	0.00	2	0	0	93826.63	0	False	False	False
4	850	43	2	125510.82	1	1	1	79084.10	0	False	True	False

Figure 3: Encoded Dataset after One-Hot Encoding

Categorical variables such as Geography and Gender were converted into numerical format using One-Hot Encoding. New binary columns including Geography_Germany, Geography_Spain, and Gender_Male were created. This transformation enabled the machine learning model to process categorical information effectively.

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6. EXPLORATORY DATA ANALYSIS

EDA was conducted to identify patterns and trends associated with customer churn.

6.1 Customer Churn Distribution

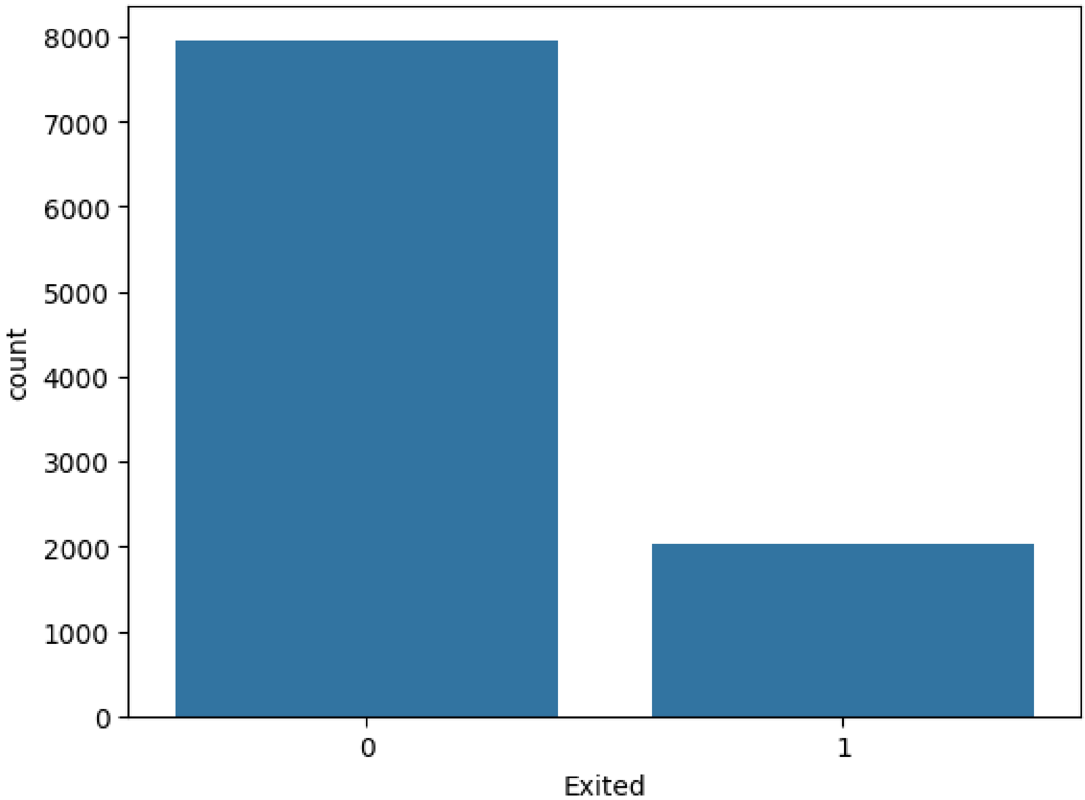

Figure 4: Distribution of Customer Churn

Interpretation

The distribution of the target variable (Exited) indicates that the dataset is imbalanced. Approximately 80% of customers remained with the bank (Exited = 0), while only about 20% of customers left the bank (Exited = 1). This suggests that customer churn occurs less frequently than customer retention. The imbalance highlights the importance of selecting robust machine learning algorithms capable of handling skewed class distributions. Understanding this distribution is crucial because churned customers represent a significant financial risk to the bank despite their smaller proportion.

6.2 Gender-Based Churn Analysis

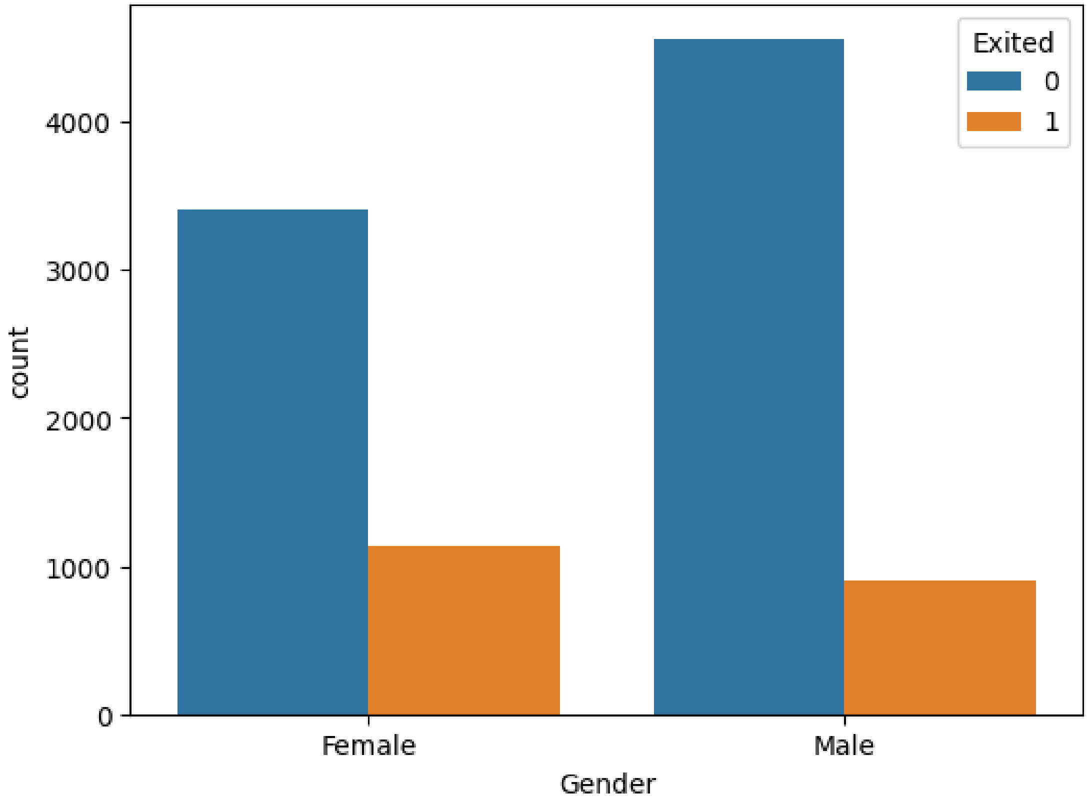

Figure 5: Customer Churn by Gender

Interpretation

The gender-wise churn analysis indicates that female customers exhibit a higher churn rate compared to male customers. Although the bank has a larger number of male customers overall, a greater proportion of female customers tend to leave the bank. This suggests that gender may have a moderate influence on customer retention and should be considered during churn prediction modeling.

6.3 Geography-Based Churn Analysis

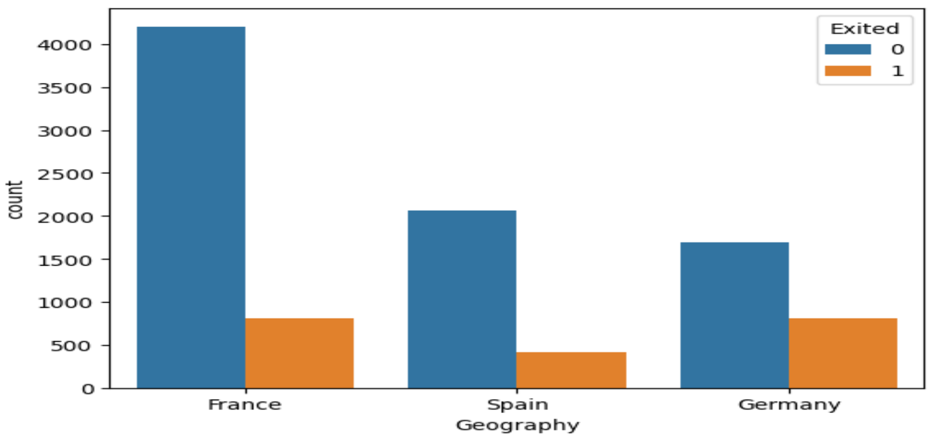

Figure 6: Customer Churn by Geography

Interpretation

The geographical analysis reveals significant differences in customer churn across regions. Germany exhibits the highest churn rate among the three countries, indicating that customers in Germany are more likely to leave the bank compared to customers in France and Spain. France has the largest customer base and the highest number of retained customers, while Spain shows the lowest churn count. These findings suggest that geographical location plays an important role in customer retention and should be considered as a key predictive feature in churn prediction models.

6.4 Age Distribution Analysis

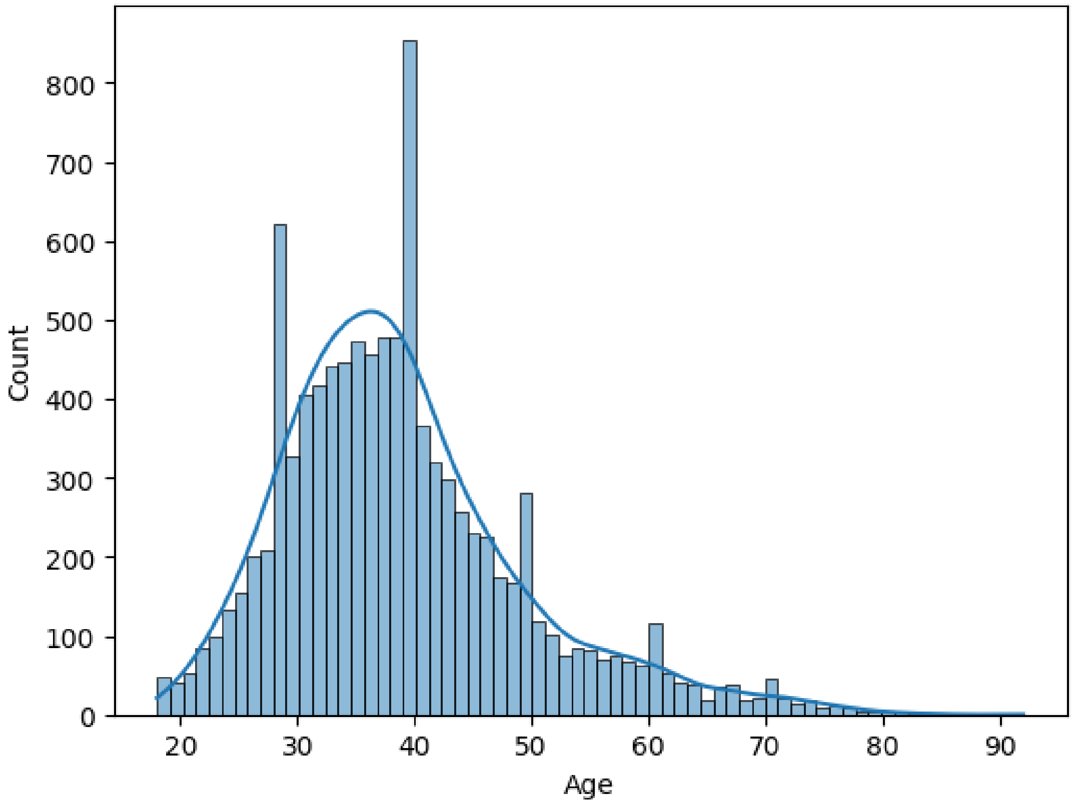

Figure 7: Distribution of Customer Age

Interpretation

The histogram illustrates the distribution of customer ages within the dataset. Most customers are concentrated between the ages of 30 and 45 years, indicating that middle-aged individuals represent the largest customer segment. The distribution is slightly positively skewed, with fewer customers in older age groups. The presence of customers across a wide age range suggests that age may influence banking behavior and customer retention. Subsequent feature importance analysis confirmed that age is one of the most significant predictors of customer churn.

6.5 Correlation Analysis

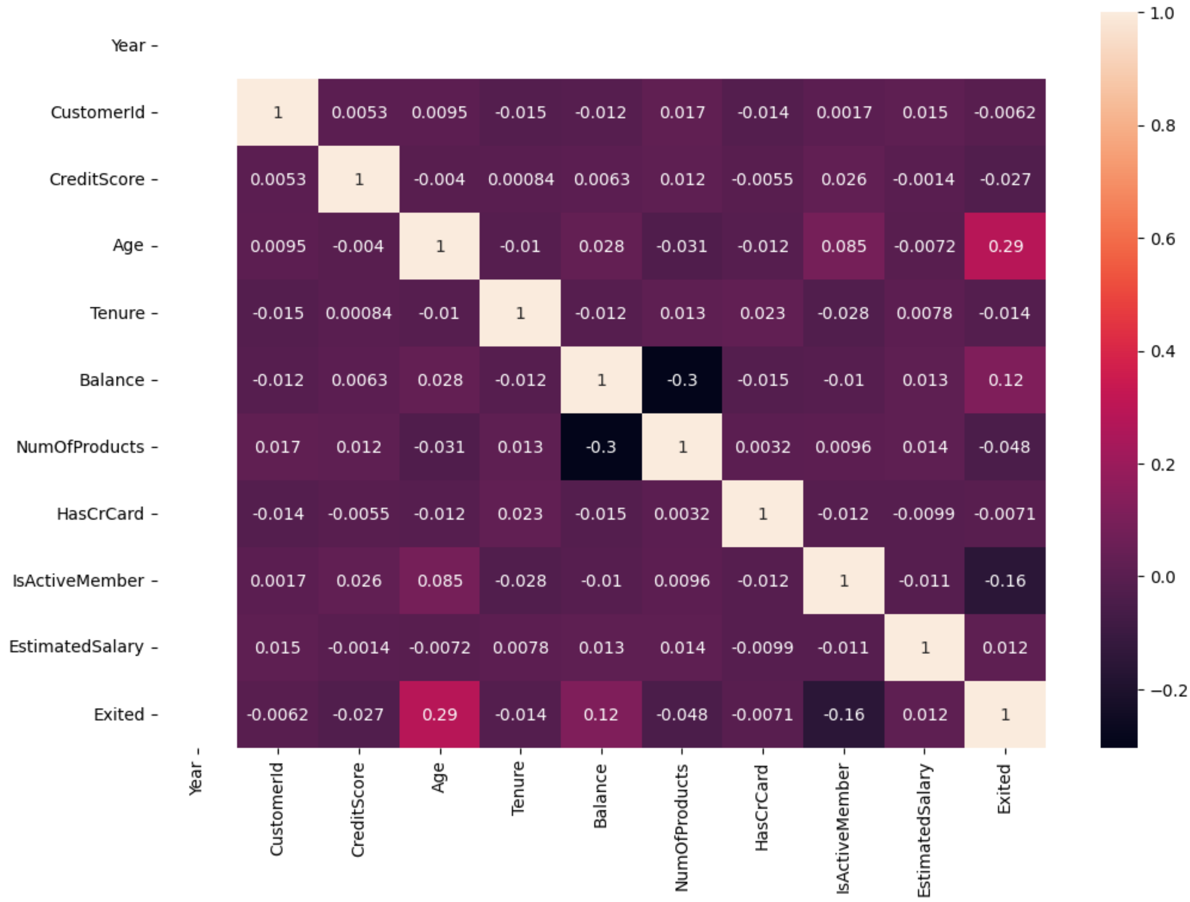

Figure 7: Correlation Matrix of Features

Interpretation

The correlation matrix was used to examine the relationships among numerical variables in the dataset. The heatmap indicates that most variables have weak correlations with one another, suggesting a low risk of multicollinearity. Among the features, Age shows the strongest positive correlation with customer churn (0.29), indicating that older customers are more likely to leave the bank. Balance also exhibits a positive relationship with churn (0.12), whereas IsActiveMember demonstrates a negative correlation (-0.16), implying that active customers are less likely to churn. These findings provide valuable insights into the factors influencing customer retention and support the effectiveness of machine learning-based churn prediction.

7. FEATURE ENGINEERING

Feature engineering improves model performance by transforming raw data into meaningful variables.

Activities performed:

• Feature selection

• One-Hot Encoding

• Data transformation

• Dataset preparation

The resulting dataset became suitable for machine learning algorithms.

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8. MODEL DEVELOPMENT

Two machine learning algorithms were implemented.

8.1 Logistic Regression

Logistic Regression served as the baseline classification model.

Advantages

• Fast training

• Easy interpretation

• Suitable for binary classification

Accuracy

80.70%

8.2 Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines multiple decision trees.

Advantages

• Higher predictive accuracy

• Better generalization

• Reduced overfitting

Accuracy

86.45%

Random Forest outperformed Logistic Regression and was selected as the final model.

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9. MODEL EVALUATION

Accuracy Comparison

Model	**Accuracy**
Logistic Regression	80.70%
Random Forest	86.45%

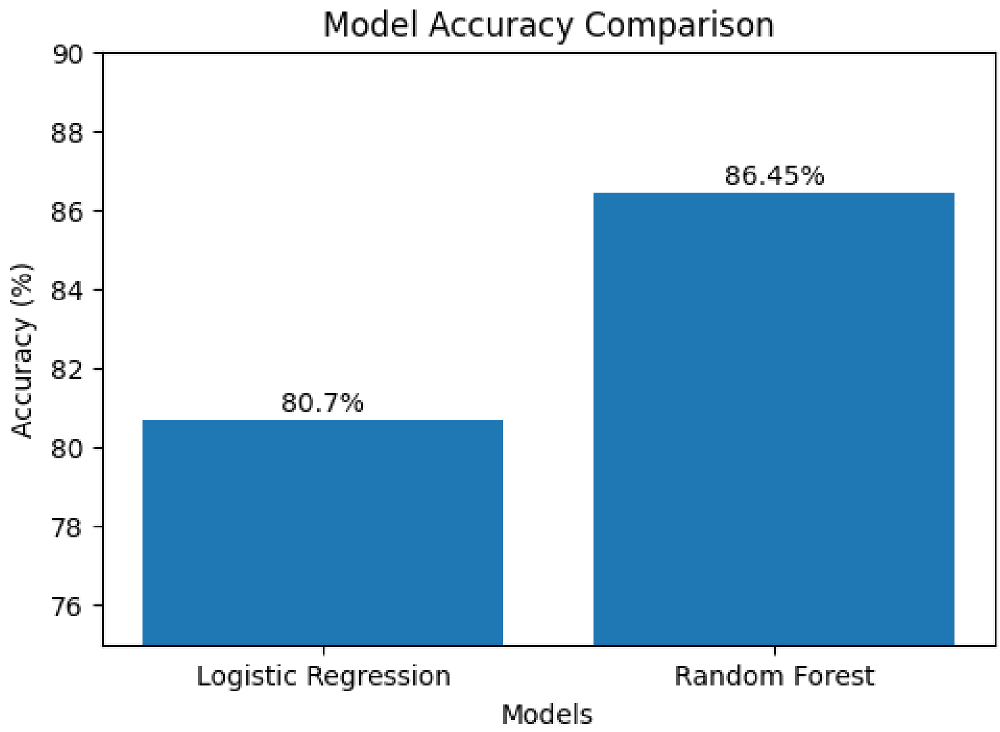

Figure 10: Accuracy Comparison of Machine Learning Models

Interpretation

the performance comparison between Logistic Regression and Random Forest algorithms used for customer churn prediction. Logistic Regression achieved an accuracy of 80.70%, whereas Random Forest achieved an accuracy of 86.45%. The higher accuracy of the Random Forest model indicates its superior capability in identifying complex patterns within customer data. Therefore, Random Forest was selected as the final predictive model for this study.

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10. FEATURE IMPORTANCE ANALYSIS

Feature importance analysis was performed using the Random Forest model.

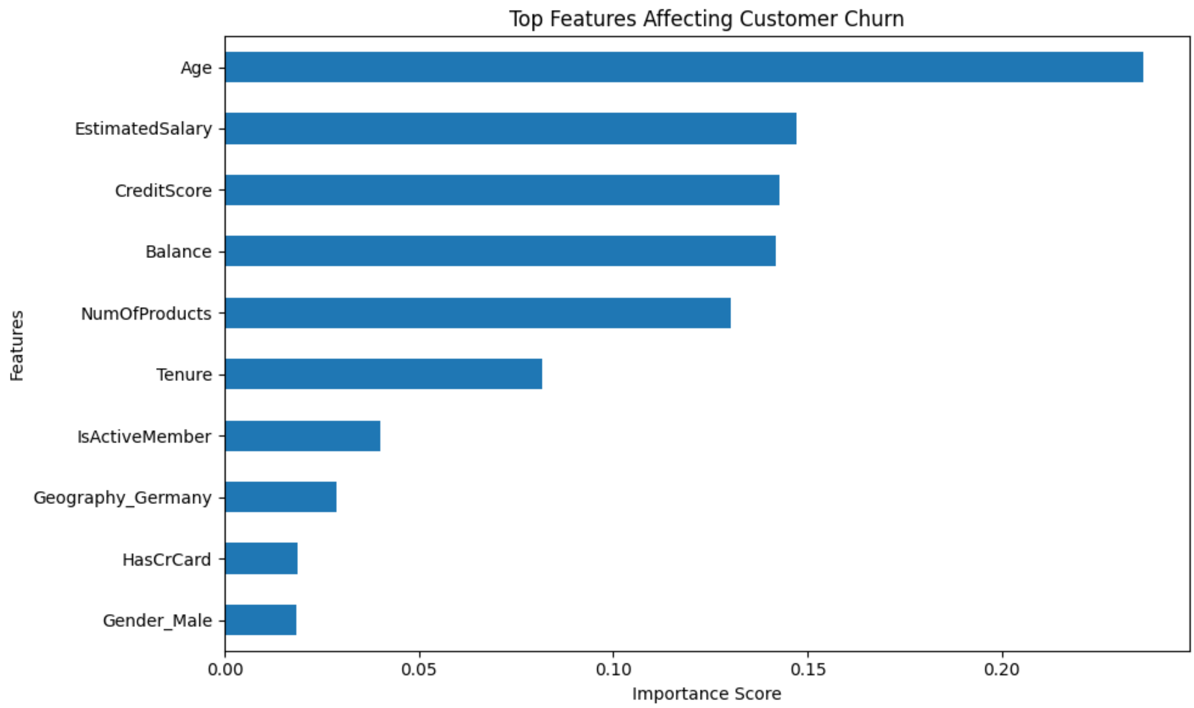

Figure 11: Top Features Affecting Customer Churn

The feature importance analysis was conducted using the Random Forest algorithm. The results indicate that Age is the most influential factor affecting customer churn, followed by Estimated Salary, Credit Score, Balance, and Number of Products. These variables play a significant role in determining whether a customer is likely to leave the bank. Understanding these key factors can help financial institutions develop targeted retention strategies and improve customer satisfaction**.**

Top Influencing Factors

1. Age
2. Estimated Salary
3. Credit Score
4. Balance
5. Number of Products
6. Tenure
7. Active Membership Status

Interpretation

Age emerged as the most influential predictor of customer churn.

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11. CHURN RISK SCORING

The final model generates probability scores indicating the likelihood of customer churn.

Examples:

Customer	Churn Probability	Risk Category
Customer A	8%	Low Risk
Customer B	22%	Medium Risk
Customer C	89%	High Risk

The Random Forest model generates churn probability scores for individual customers. These scores help classify customers into Low, Medium, and High-Risk categories. Customers with higher churn probabilities should be prioritized for retention campaigns, personalized offers, and customer engagement strategies. This risk-scoring approach enables banks to take proactive actions and reduce customer attrition.

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12. RESULTS AND FINDINGS

The machine learning models were trained and evaluated to predict customer churn in the banking sector. The Random Forest classifier achieved the highest accuracy of 86.45%, outperforming Logistic Regression, which achieved 80.70% accuracy.

The exploratory data analysis revealed several important patterns:

- Customer age showed a strong relationship with churn behavior.
- Customers with higher account balances were more likely to leave the bank.
- Active members demonstrated lower churn rates compared to inactive customers.
- Geography influenced churn, with some regions showing higher customer attrition.
- Credit score and number of products also contributed to churn prediction.

The feature importance analysis identified Age, Estimated Salary, Credit Score, Balance, and Number of Products as the most influential factors affecting customer churn.

These findings indicate that customer behavior and demographic characteristics play a significant role in predicting churn and can help banks design effective retention strategies.

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13. BUSINESS RECOMMENDATIONS

The following recommendations can help banks reduce customer churn:

1. Develop targeted retention programs for high-risk customers.
2. Improve customer engagement through personalized communication.
3. Offer loyalty rewards and special benefits to long-term customers.

4. Focus on inactive customers and encourage greater service usage.
5. Monitor customer complaints and resolve issues quickly.
6. Provide customized financial products based on customer profiles.
7. Use predictive analytics regularly to identify customers likely to churn.

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14. CONCLUSION

This study successfully developed a machine learning-based customer churn prediction system for the banking sector. The dataset was preprocessed, analyzed, and used to train multiple classification models.

Among the evaluated models, Random Forest achieved the best performance with an accuracy of 86.45%. The model effectively identified customers who were likely to leave the bank and highlighted the key factors influencing churn behavior.

The proposed solution can help financial institutions improve customer retention, reduce revenue loss, and support data-driven decision-making.

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15. FUTURE SCOPE

Future enhancements of this project may include:

- Using larger and real-time banking datasets.
- Implementing advanced machine learning algorithms such as XGBoost and LightGBM.
- Deploying the model as a web application for real-time prediction.
- Integrating customer feedback and transaction history into the model.
- Applying deep learning techniques for improved predictive performance.
- Developing automated retention recommendation systems.

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